

ESP32 Low-Power Setup Summary

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1. LDO Requirements (MCP1700-3.3)

- Use MCP1700-3.3 as the voltage regulator.
- Ultra-low quiescent current ($\sim 1.6 \mu\text{A}$ typical).
- Input voltage up to 6.0V. Suitable for Li-ion (3.0–4.2V).
- Max output current 250 mA supports ESP32 WiFi bursts.
- Dropout voltage $\sim 178 \text{ mV}$ at 100 mA.
- Required capacitors:
 - 10 μF ceramic at input.
 - 10 μF ceramic at output.
 - (Optional) 470 μF electrolytic near ESP32 3.3V pin for WiFi spikes.

2. Power Architecture

Battery $\rightarrow$ MCP1700-3.3 $\rightarrow$ ESP32 3V3

All grounds common.

Hall sensor and pull-up resistors powered at 3.3V.

3. Programming the Bare ESP32-WROOM-32U

Use an ESP32 DevKit as USB-to-UART adapter.

Wire:

- DevKit TX $\rightarrow$ New module RX
- DevKit RX $\rightarrow$ New module TX
- DevKit 3V3 $\rightarrow$ New module 3V3
- DevKit GND $\rightarrow$ New module GND

- DevKit GPIO0 → New module GPIO0 (BOOT)
- DevKit EN → New module EN (RESET)

Hold DevKit's onboard ESP32 in reset (EN tied to GND or hold reset button) so it does not interfere.

To program:

- Pull GPIO0 LOW during reset to enter bootloader.
- Upload from Arduino IDE as normal.
- Remove GPIO0 LOW for normal boot.

4. Low-Power Usage Strategy

- Use deep sleep for lowest power (~20 μ A).
- Wake on GPIO interrupt from hall sensor (GPIO34).
- Keep WiFi off except for brief transmission windows.
- Avoid DevKit for deployment; bare module + MCP1700 yields long battery life.